

GAURAV KUMAR VERMA

Software Engineer · AI/ML Builder · Competitive Programmer
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ABOUT

B.Tech Computer Science student at IIIT Noida (CGPA 7.9) with a track record of building complete, working AI-powered products — from a brain tumor classifier with ~98.66% accuracy and an offline NLP desktop automation system, to a deployed healthcare AI platform and a real-time pollution monitoring API with ML forecasting. Comfortable across the full stack: React/Next.js frontends, FastAPI backends, ML pipelines (Scikit-learn, TensorFlow/Keras), and systems-level Python automation. Strong problem-solver with C++ and competitive programming experience. Oracle AI Foundations certified. Looking to contribute to teams building impactful, well-engineered software.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, C
Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, Vite
Backend & APIs: FastAPI, Node.js, Firebase, REST APIs, Vercel AI SDK
AI / ML: Scikit-learn, TensorFlow/Keras, EfficientNetV2, Ollama (local LLMs), Tesseract.js (OCR), Grad-CAM, Prompt Engineering
Data: Pandas, NumPy, Time-series Forecasting, Clustering, Geospatial Visualization
Tools & Platforms: Git, GitHub, Vercel, VS Code, SQL
CS Fundamentals: Data Structures & Algorithms, OOP, System Design, Networking

PROJECTS

Brain Tumor Classification – Robust Deep Learning 2025

Python · TensorFlow/Keras · EfficientNetV2-S · Grad-CAM

- Trained EfficientNetV2-S on multi-class MRI data for brain tumor classification, achieving ~98.66% test accuracy across 4 classes.
- Boosted robustness by training on combined clean + noise-augmented datasets; applied Grad-CAM to generate visual activation maps for clinical interpretability.

ATLAS – Local AI Desktop Automation System 2025

Python · Ollama (local LLM) · NLP/Regex pipeline · Desktop APIs

- Built a fully offline NLP-to-action engine: rule-based regex classifier handles known commands (~85-90% accuracy); local Ollama LLM resolves ambiguous inputs as fallback — zero cloud dependency.
- Implemented risk-gated execution layer (validation → risk scoring → confirmation) before performing real desktop actions (app launch, file operations, etc.), with persistent session memory.

HealthSync – AI Healthcare Continuity Platform 2025

React · Vite · Tailwind · Redux Toolkit · Tesseract.js · Vercel

- Built multi-role healthcare dashboard (patient/doctor/admin) with OCR-powered prescription digitization using Tesseract.js, rule-based NLP analysis, and predictive monitoring alerts.
- Shipped as a live Vercel prototype in a team; awarded Certificate of Excellence at Hackoverflow 9.0.

Delhi Ward Pollution Monitor (DWLP) 2025

React · FastAPI · Scikit-learn · Geospatial libraries

- Developed full-stack platform for real-time ward-level AQI monitoring across Delhi with ML-driven forecasting (time-series regression) and clustering-based policy recommendations.

- FastAPI backend ingests real Delhi AQI datasets, runs forecasting models, and serves geospatial visualizations to the React frontend.

CodeVibe – AI Prompt Engineering Platform

2025

Next.js · Firebase Auth · Vercel AI SDK · OpenAI / Anthropic APIs

- Built a Next.js platform for structured AI prompt generation across SDLC phases with BYOK (Bring Your Own Key) integration — supporting OpenAI and Anthropic APIs via browser-to-API communication.
- Implemented Firebase Authentication and multi-provider API switching with zero server-side key storage.

EDUCATION

B.Tech – Computer Science & Engineering

May 2024 – May 2028

Jaypee Institute of Information Technology, Noida · CGPA: 7.9 · Ranked Top 10 – College HackerRank Contest

Higher Secondary – Non-Medical (Science)

Apr 2024

St. Joseph's Convent School, Faridabad · 94%

CERTIFICATIONS & ACHIEVEMENTS

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- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
 - Cisco Certified Networking Basics
 - Certificate of Excellence – Hackoverflow 9.0 (HealthSync project, competitive hackathon)
 - Certificate of Appreciation – Paranox 2.0 · Certificate of Appreciation – HackVision

LEADERSHIP & VOLUNTEER

Technical Head – National Service Scheme (NSS), IIIT Noida

Oct 2025 – Present

- Led technical operations for community outreach events promoting environmental and social awareness.
- Tutored students in technical subjects; produced informational content on environmental issues for community distribution.